M&AE 5070 - Dynamics of Flight Vehicles

Chapter 8 - Lateral Performances

In this chapter, we will build on the steady turn results established previously to investigate aircraft performances during turns, and more specifically, determining the maximum rotation rate, $\Omega_{t,\max}$, and minimum turn radius, $R_{t,\min}$, that an aircraft can achieve during a turn.

Assumptions

- For simplicity, we will assume that **the aircraft engines are not limiting**. Note that a similar process can be followed to determine turn performances when the engines become the limiting factor.

From last chapter: turn at constant altitude necessarily needs an increase in thrust

Limiting factor here : related to aerodynamics, and more specifically sustentation / stall

- steady turns [equations from previous chapter apply]
- no side-slip β (turn made with both δ_ℓ and δ_n)
- constant altitude : $\gamma = 0$ and ρ = constant
- constant velocity [typical + simplifies greatly comparison between straight flight and subsequent turn]

Maximum rotation rate $\Omega_{t,\max}$

The rotation rate Ω_t can be expressed as a function of the load factor during the turn, $n_{z,t}$ (from previous chapter):

$$\Omega_t = \frac{g}{V}\sqrt{n_{z,t}^2 - 1}$$

$n_{z,t}$ can be expressed as the ratio of C_z during the turn and during the straight flight before starting the turn:

$$n_{z,t} = \frac{C_{z,t}}{C_z}$$

C_z during turn

C_z during straight flight before turn

Clearly, $n_{z,t}$ depends on the initial straight flight conditions, more specifically on its velocity:

S before turn : $mg = \frac{1}{2}\rho S V^2 C_z$

S during turn : $n_{z,t} = \frac{1}{2}\rho S V^2 \frac{C_{z_t}}{mg}$

Ω_t therefore depends on two main quantities: the lift coefficient during the turn, and the velocity at which

the turn is made. Since we are looking for the maximum rate of turn $\Omega_{t,\max}$:

- the lift coefficient during the turn should be maximized:
- what to do with the velocity is less clear, since it appears both at the denominator and at the numerator.

$$\Omega = \frac{g}{V}\sqrt{n_{z,t}^2 - 1}$$

$n_{z,t}$: increases when V increases

$\frac{g}{V}$: decreases when V increases

As usual, we can attempt to find an optimal value for V by differentiating Ω_t with respect to the velocity and equating the resulting expression to zero:

Don't want to deal with $\sqrt{}$ $\Rightarrow$ consider Ω^2 and V^2 instead of Ω and V

$$\Omega^2 = \frac{g^2}{V^2}\left(n_{z,t}^2 - 1\right) = g^2\left(K^2V^2 - \frac{1}{V^2}\right) \quad \text{with} \quad K = \frac{1}{2}\rho S\frac{C_{z,t}}{mg}$$

$$\frac{d\Omega^2}{dV^2} = g^2\left(K^2 + \frac{1}{V^4}\right)$$

ALWAYS >0, NO OPTIMUM!
The higher the velocity, the higher the rate of rotation Ω_t
↳ the load factor contribution dominates

Since we can't just let V take arbitrarily large values, we need to consider additional physical constraints to determine $\Omega_{t,\max}$:

ignore here:
- Constraint on max thrust that engines can provide, F_{max}
  ↳ soft constraint : attempting to exceed will result in loss of altitude
- Constraint on the structural integrity of the aircraft

There are 2 characteristic load factors of relevance here:

- Maximum load factor that does not lead to permanent structural deformation or damage

$$n_{z_s} \approx 2.5 \text{ [civil]}$$
$$\approx 10 \text{ [military]}$$

- Maximum load factor leading to catastrophic structural failure [extreme loading]

$$n_{z_e} = 1.5\, n_{z_s}$$

$\Rightarrow$ $n_{z,t}$ during turn CANNOT exceed n_{z_s}

- At low velocity, the load factor is small ($< n_{z_s}$) and increases as the velocity increases
- When the velocity increases, the rotation rate increases

For a certain velocity V^*:

- the load factor will reach the structural load factor n_{z_s}
- the rotation rate will reach its maximum structurally allowable value

$\Rightarrow$ that velocity is called CORNER Velocity.

Let's find this corner velocity V^*. We can write the following relationships:

when $\Omega_t = \Omega_{t,max}$: $n_{z,t} = n_{z_s} = \frac{C_{z,t}}{C_z}$

needs to be maximum for Ω_t to be maximum for a given velocity

$C_{z,t} = C_{z_{max}}$

n_{z_s}: limit for permanent structural damage

C_z during straight flight associated with corner velocity V^*

$$C_z = \frac{mg}{\frac{1}{2}\rho S V^{*2}}$$

S during turn : $n_{z_s} mg = \frac{1}{2}\rho S V^{*2} C_{z_{max}}$

S on straight flight AT STALL CONDITIONS $mg = \frac{1}{2}\rho S V_{STALL}^2 C_{z_{max}}$

Taking ratio between the 2 : $n_{z_s} = \frac{V^{*2}}{V_{STALL}^2}$

OR $\boxed{V^* = \sqrt{n_{z_s}}\, V_{STALL}}$

At the corner velocity, rotation rate becomes

$$\Omega_t = \Omega_{t,max} = \frac{g}{V^*}\sqrt{n_{z_s}^2 - 1} = \frac{g}{\sqrt{n_{z_s}}\, V_{STALL}}\sqrt{n_{z_s}^2 - 1}$$

$$\boxed{\Omega_{t,max} = \frac{g}{V_{STALL}}\sqrt{n_{z_s} - \frac{1}{n_{z_s}}}}$$

For example, for a transport aircraft, the corner speed that will allow the aircraft to reach max rotation rate is $\sqrt{2.5} = 1.6$ times its stall velocity (at given altitude). For military aircrafts, it would be $\sqrt{10} \approx 3.2$ times V_{STALL}

Minimum rotation radius $R_{t,min}$

The procedure to obtain an expression for $R_{t,min}$ is very similar to that of the maximum rotation rate. We start from the definition of the (inverse of) rotation radius R_t:

$$\frac{1}{R_t} = \frac{\Omega_t}{V} \tag{1}$$

Minimizing R is equivalent therefore to maximizing $\frac{\Omega_t}{V}$. As for the rotation rate, it turns out that there is no minimum, and that again, we need to place a constraint on the load factor, which cannot be above $n_{z,s}$. A bit of algebra shows that in this case, the minimum rotation radius is such that:

$$\frac{1}{R_{t_{min}}} = \frac{\Omega_{t,max}}{\sqrt{n_{z_s}}\, V_{STALL}}$$

The optimal velocity for this case is still the "corner speed" defined above.

Order of magnitude

Aircraft	V_{S_s} kt	V_{S_s} m/s	n_{z_s}	$\sqrt{n_{z_s} - \frac{1}{n_{z_s}}}$	$\Omega_{t,\max}$ rad/s	$\Omega_{t,\max}$ $^o/s$	$\Omega_{t,\max}$ rpm	$V^\star$ kt
Transport, high-lift systems	105	54	2.5	1.45	0.263	15	2.5	166
Transport, no high-lift	150	77	2.5	1.45	0.185	10.5	1.75	237
Military	140	72	≈ 10	3.15	0.43	24.4	4	448

Table 1: Typical rotation rate for different aircrafts. The transport aircraft used as example is the twin-engine, short-to-mid-haul aircraft used throughout the case studies (≈ 250 passengers), either with or without high-lift systems deployed. The military aircraft data are from a mid-90s typical single-seat, single-engine fighter jet. Ω for a fighter jet is much larger (not surprising), and the optimal speed $V^\star$ is also much larger.

Application: A fighter jet pilot evaluates the turn performances of her aircraft. She measures her maximum rotation rate at an altitude of 11 km: $\Omega_{t,\max}|_{11km}$. She then increases her altitude to 15.5 km, and is about to measure her maximum rotation rate again. Before the test, she quickly runs some numbers and estimates her maximum rotation rate $\Omega_{t,\max}|_{15.5km}$ around:

You can assume that the mass remains constant.

1. 0.7 $\Omega_{t,\max}|_{11km}$
2. 0.5 $\Omega_{t,\max}|_{11km}$
3. 1.4 $\Omega_{t,\max}|_{11km}$

4. $2\ \Omega_{\ell,\max}|_{11km}$
5. We are missing data to evaluate the new rotation rate.

Solution: 1. The maximum rotation rate at the limit load factor is given by:

$$\Omega_{\ell,\max} = \frac{g}{V_{S,s}}\sqrt{n_{z_s} - \frac{1}{n_{z_s}}} = \frac{g}{V^\star}\sqrt{n_{z_s}^2 - 1} \tag{2}$$

where n_{z_s} is the structural load factor, fixed for the aircraft and independent of altitude. $V_{S,s}$ is the stall velocity on a straight flight at the altitude of interest. Since $C_{z_{\max}}$ is only weakly dependent on the altitude, we can write that $\rho V_{S,s}$ is constant. Going up by 4.5 km, the stall velocity will therefore be multiplied by $\sqrt{2}$, so the maximum rotation rate will be divided by $\sqrt{2}$. Performances decrease when altitude increases!